DESIGN AND EVALUATION OF AN AUGMENTED REALITY STORYBOOK FOR PRIMARY LEARNERS' PLANT GROWTH UNDERSTANDING & STEWARDSHIP

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Young learners often find it difficult to understand scientific concepts related to plant growth and environmental influence when learning is limited to static images in textbooks and traditional classroom explanations. In early primary education, students benefit from interactive and visual learning experiences that help them connect abstract concepts with real-world phenomena. Although augmented reality (AR) has been widely explored as a tool for interactive education, many existing AR systems mainly focus on visual presentation rather than creating dynamic simulations that respond to real-world environmental conditions. To address this limitation, this research proposes a context-aware augmented reality plant simulation system designed to demonstrate how environmental factors influence plant growth. The proposed system integrates real-world environmental sensing with a virtual simulation engine that allows a plant model to grow dynamically within an AR environment. The system collects environmental data such as light intensity and device motion through mobile device sensors and translates this information into simulation parameters that influence plant behavior. The architecture consists of several interconnected modules including environmental sensing, sensor processing and fusion, a context-aware plant growth simulation engine, interactive environmental feedback mechanisms, and AR visualization. The system will be implemented using Unity 3D with AR Foundation and ARCore, enabling real-time rendering of the virtual plant within the user's physical environment. Environmental interaction features such as using the mobile device flashlight as virtual sunlight and motion sensors to simulate wind effects will strengthen the connection between the real environment and the virtual simulation. The proposed system will be evaluated through experimental testing to analyze system responsiveness, simulation realism, and interaction effectiveness. Expected outcomes include improved understanding of environmental factors affecting plant growth, enhanced learner engagement through interactive AR experiences, and the demonstration of a scalable framework for integrating environmental sensing with simulation-based AR learning environments. From an educational perspective, the research contributes to more interactive science learning experiences for young learners, while from a technological perspective it demonstrates how context-aware simulation can extend the capabilities of mobile augmented reality systems.

Keywords: *Augmented Reality, Context-Aware Simulation, Virtual Plant Growth, Environmental Sensing, Interactive Learning, Mobile AR Systems.*

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
AR	Augmented Reality
ARCore	Augmented Reality Core
ARKit	Augmented Reality Kit
CHI	Conference on Human Factors in Computing Systems
MR	Mixed Reality
UI	User Interface
VR	Virtual Reality

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Appendix A – Survey Instruments

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1. INTRODUCTION

1.1 Background and Context

In early primary education, young learners often find it difficult to understand scientific concepts when learning is limited to static textbook illustrations and verbal explanations. This challenge is particularly noticeable in science subjects such as plant growth and environmental interactions, where many processes occur gradually over time and cannot be easily visualized through traditional learning materials. As a result, students at the early primary level may struggle to connect theoretical knowledge with real-world observations. Interactive learning tools that allow students to visualize and explore scientific concepts can therefore play an important role in improving understanding and engagement [1], [15].

Augmented Reality (AR) has emerged as a promising technology for creating immersive learning environments that combine digital content with the physical world. AR systems overlay virtual objects onto real-world environments, allowing learners to interact with digital simulations while observing their physical surroundings. Previous studies have shown that AR-based learning tools can enhance student motivation, engagement, and conceptual understanding by transforming abstract concepts into interactive experiences [1], [13].

In the context of this research, the overall system is designed as a handheld AR storybook that supports Cambridge Primary Science Grade 1 learners in Sri Lankan international schools. The storybook combines printed pages with augmented reality content, allowing students to interact with educational simulations through a mobile device [1]. By pointing the device camera at the storybook pages, learners can observe animated and interactive AR elements that help explain scientific concepts in a more engaging and intuitive manner [15].

Modern mobile devices include various sensors such as cameras, gyroscopes, accelerometers, and ambient light sensors. These sensors allow mobile applications to capture environmental information and user movements in real time [5]. In AR systems, such sensor data can be used to align virtual objects

with the physical environment and enable dynamic interaction between real and virtual elements. Sensor fusion techniques, which combine data from multiple sensors, have been shown to improve motion tracking accuracy and stability in mobile AR systems [4].

Another important area of research is the simulation of natural systems within digital environments. Procedural modeling techniques have been widely used to simulate natural phenomena such as plant growth, environmental changes, and biological structures. These models use mathematical algorithms and predefined parameters to generate realistic plant shapes and development patterns over time. For example, procedural techniques have been developed to simulate leaf growth and venation patterns using biologically inspired mathematical models [7].

However, many existing plant simulation models operate using fixed parameters and do not respond to real-world environmental conditions. In many AR applications, virtual plants or environmental elements are displayed mainly for visualization purposes without dynamically reacting to real-world environmental data. As a result, the interaction between the physical environment and the virtual simulation remains limited [10].

Context-aware computing offers a possible solution to this limitation. Context-aware AR systems use environmental information and user interactions to adapt digital content dynamically. By integrating environmental sensing, such systems can respond to factors such as lighting conditions, device movement, and user behavior to provide more interactive and adaptive experiences. Research has highlighted the potential of context-aware AR applications to create richer and more responsive interactions between digital simulations and real-world environments [2], [12].

Based on these technological advancements, this research focuses on developing a context-aware simulation engine for the AR storybook system that connects real-world environmental data with virtual plant growth simulations. In this approach, environmental information such as light intensity and device movement will influence the behavior and growth of a virtual plant within the AR environment. For example, the system will analyze the surrounding lighting conditions to determine whether the environment is bright or dark.

If the environment is too dark, the device flashlight can act as a virtual grow light to support the growth of the virtual plant. Additionally, gyroscope data can be used to simulate environmental effects such as wind or water interactions.

By integrating environmental sensing, procedural plant simulation, and interactive AR storytelling, the proposed system aims to create a learning experience that allows Grade 1 students to explore plant growth in an engaging and interactive way. Through this approach, learners can observe how environmental conditions influence plant development, helping them understand scientific concepts through direct interaction and experimentation within the AR storybook environment.

1.2 Literature Review

Augmented Reality (AR) combines digital objects with the real world, relying on device sensors to sense the environment [13]. Modern mobile AR systems use cameras, light sensors, gyroscopes, and more to capture environmental context in real time [4], [5]. Researchers have shown that context-aware AR where applications adjust to lighting, user position, and other real-world conditions can greatly enhance immersion and usability [2], [12]. For example, Scargill et al. (2021) emphasize that truly context-aware AR should use not only the device's own sensors but also data from wearables or IoT devices to enrich environmental understanding [2]. By fusing these data streams, AR applications can better understand ambient light, temperature, spatial layout, and other context, enabling more realistic AR interactions [4], [2]. However, many existing AR apps only use camera input and basic tracking; they often do not fully incorporate all environmental data into their simulations. This indicates a need for AR systems that more deeply integrate live sensor data into the simulation engine, beyond just using it for visual adjustments [10], [11].

1.2.1 Context-Aware Environmental Sensing in Mobile and AR Systems

Context-aware sensing in mobile AR involves using device sensors to capture real-world conditions and feeding that data into the AR application [2]. These sensors allow AR systems to observe the surrounding environment and adjust virtual elements based on real-world conditions. As a result, context-aware sensing plays an important role in making AR experiences more realistic and responsive.

In mobile AR systems, environmental sensing is commonly used to monitor factors such as lighting conditions, device orientation, motion, and user interaction [4], [5]. The information collected from these sensors is then used to adjust how virtual objects are displayed or behave in the AR environment. For example, modern AR platforms such as ARCore and ARKit include light estimation techniques that analyze the camera image to understand the brightness and color of the surrounding environment. This information allows virtual objects to adjust their brightness, shading, and shadows so that they appear consistent with the real-world lighting conditions [3]. By matching the illumination of the virtual and physical environments, AR applications can create a more believable visual experience.

In practice, this means monitoring the environment's lighting, the device's orientation, and user actions to adapt the virtual content. For example, advanced AR frameworks can estimate the scene's illumination by analyzing the camera image, and then adjust the brightness and shading of virtual objects accordingly. In addition to lighting detection, mobile AR systems rely heavily on motion sensors to track the position and orientation of the device. Accurate tracking is necessary to ensure that virtual objects remain stable and correctly aligned with the real-world environment. Research by Gustafsson et al. (2008) showed that combining accelerometer and gyroscope data with camera-based tracking significantly improves the accuracy of device pose estimation [4]. Their work demonstrated that using a nonlinear Kalman filter to fuse multiple sensor inputs allows the system to estimate device motion more accurately than using a single sensor alone. This type of sensor fusion is widely used in AR systems to maintain stable tracking of virtual objects even when the device moves quickly or when visual features are limited.

Gustafsson et al. (2008) demonstrated that fusing accelerometer and gyroscope data with camera tracking via a nonlinear Kalman filter yields very accurate device pose estimates [4]. This kind of sensor fusion is a form of context awareness that improves AR tracking robustness. Scargill et al. argue that adding data from wearables and IoT devices would push context awareness further [2]. In addition to device sensors, researchers have suggested expanding context-aware sensing by incorporating data from external sources. Scargill et al. suggest that integrating information from wearable devices and Internet of Things (IoT) sensors could further improve environmental awareness in AR systems [2]. For example, environmental sensors placed in smart environments could provide additional information such as temperature, humidity, or ambient light levels. When combined with mobile device sensors, this additional data could help AR systems better understand real-world conditions and support more advanced context-aware applications.

Therefore, there is still an opportunity to explore more advanced uses of environmental sensing in AR systems. Instead of using sensor data only for rendering adjustments, future systems can use real-world environmental information to influence virtual processes and behaviors. Integrating environmental sensing with simulation engines would allow virtual environments to react dynamically to real-world conditions, creating more interactive and meaningful AR experiences.

In summary, context-aware sensing in AR uses multiple sensors to make the system aware of real-world conditions; however, many AR systems today still only use this data for basic tracking and lighting rather than driving complex simulations.

1.2.2 Simulation of Natural Systems and Procedural Plant Growth

Procedural simulation is widely used to model natural phenomena like plant growth [7]. Instead of manually designing every detail of a plant model, procedural techniques use mathematical rules and algorithms to generate plant structures automatically. This approach allows complex natural forms to be created efficiently and enables simulations where plants can grow, change, or respond to different conditions over time.

Researchers have developed algorithms that mimic biological processes to generate realistic plant shapes. Many researchers have studied computational models that attempt to replicate the biological processes involved in plant growth. For example, Alsweis and Deussen (2015) proposed a procedural method for simulating leaf development based on biological growth mechanisms. In their model, a leaf is treated as an expanding surface where growth is influenced by hormone flows such as auxin. The algorithm simulates how cells expand and how veins develop across the leaf surface, which results in realistic leaf shapes and venation patterns. Their system is capable of automatically extracting the outline of a leaf and generating a detailed growth structure using mathematical rules that represent biological processes [7]. This type of modeling shows how procedural simulation can reproduce complex natural patterns using relatively simple computational rules.

For instance, Alsweis and Deussen (2015) presented a model where a leaf is treated as an expanding fluid influenced by "hormone" (auxin) flows, resulting in realistic leaf contours and vein patterns [7]. Their system automatically extracts the leaf's outline and simulates tissue growth and venation development using this mathematical model.

In addition to leaf growth models, other studies have explored procedural methods for simulating entire plant structures such as stems, branches, and trees. One widely known approach is the use of L-systems (Lindenmayer systems), which describe plant development using sets of rewriting rules. These rules are applied repeatedly to generate branching patterns that resemble real plants. L-systems have been used extensively in computer graphics to generate trees, vines, and other vegetation structures in virtual environments [13]. Because the rules can be adjusted easily, developers can create a wide variety of plant forms while maintaining realistic biological patterns.

Another group of simulation approaches focuses on physics-based modeling. These models consider factors such as gravity, structural strength, and environmental forces when simulating plant growth. Physics-based methods are particularly useful when plants interact with environmental conditions such as wind, sunlight direction, or obstacles in the surrounding environment. By combining biological growth rules with physical simulation, these systems can create more realistic and dynamic plant behavior [8]. Despite these advancements, many existing plant growth simulations rely on predefined environmental parameters rather than real-world environmental data. In most cases, growth conditions such as light intensity, temperature, or water availability are manually set within the simulation environment before the simulation begins. Once the parameters are defined, the plant model follows a fixed growth pattern based on those values. This means that the simulation does not adapt dynamically to changes in the surrounding environment during execution.

Because of this limitation, most procedural plant models operate independently from the real-world conditions experienced by the user. For example, a virtual plant may grow according to a predefined light level even if the actual environment around the device becomes darker or brighter. As a result, the simulation may not reflect real environmental changes, reducing the realism and interactivity of the system.

This limitation creates an opportunity to improve plant simulation systems by integrating real-world environmental data into the simulation process. If procedural plant growth algorithms are connected with environmental sensing technologies, virtual plants could respond directly to real environmental conditions. For example, sensor data such as ambient light levels, device orientation, or user interaction could influence the growth speed, health, or appearance of a virtual plant.

Such an approach would allow simulation systems to move beyond static parameter settings and become truly context-aware. By linking real-world environmental sensing with procedural plant growth models, AR applications could create dynamic virtual ecosystems where plant behavior changes according to real environmental conditions. This integration of environmental sensing and procedural simulation forms an important foundation for the context-aware simulation engine proposed in this research.

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1.2.3 Sensor Fusion and Energy-Efficient Data Processing in Mobile AR

Sensor fusion plays an important role in mobile augmented reality (AR) systems because AR applications rely on multiple sensors to understand the physical environment [4]. Modern smartphones contain several sensors such as accelerometers, gyroscopes, cameras, light sensors, and GPS modules [5]. Each sensor provides a different type of information about the environment or device movement. Sensor fusion combines data from these multiple sensors to produce more accurate and reliable information about the user's surroundings.

One of the most important applications of sensor fusion in AR is device pose estimation. Pose estimation refers to determining the position and orientation of the device in three-dimensional space. Accurate pose estimation is necessary to ensure that virtual objects remain correctly aligned with the real world when the user moves the device. Gustafsson et al. (2008) demonstrated that combining inertial sensor data with camera-based tracking significantly improves the accuracy of pose estimation in AR systems. Their approach used a nonlinear Kalman filter to fuse accelerometer and gyroscope measurements with visual tracking information obtained from the camera. The results showed that this fusion method can reduce tracking errors and improve the stability of virtual object placement [4].

In addition to improving tracking accuracy, sensor fusion can help AR systems better understand environmental conditions. For example, environmental sensors such as ambient light sensors can provide information about lighting intensity in the surrounding environment. When this data is combined with camera analysis, AR applications can more accurately estimate the overall illumination of the scene. This allows the system to adjust virtual object brightness and shading so that they match real-world lighting conditions. Similarly, motion sensors such as gyroscopes and accelerometers provide continuous information about device movement, which allows the AR system to maintain stable rendering of virtual objects even when the user moves the device quickly [11].

Beyond motion and lighting detection, sensor fusion can also support more advanced interactions in AR systems. By combining data from different sensors, AR applications can recognize user actions and environmental changes more effectively. For instance,

gyroscope data can detect device tilting or rotation, while camera input can identify surfaces or objects in the environment. When these data sources are combined, the system can provide more responsive and context-aware interactions [12].

Despite the advantages of sensor fusion, mobile AR systems face several challenges related to performance and energy consumption. Processing multiple sensor streams continuously requires significant computational resources. In addition, rendering complex virtual objects and simulations can place a heavy workload on the device's processor and graphics unit. As a result, many mobile AR applications experience high energy consumption and reduced battery life.

Research has shown that energy efficiency is a critical issue for mobile AR applications. Brocanelli et al. introduced the eAR framework to address performance and energy limitations in mobile AR systems. Their work highlighted that many AR applications require intensive computation for tasks such as rendering complex 3D objects, environmental analysis, and sensor data processing. These operations can quickly drain the device's battery and negatively affect the user experience [6]. To address this issue, the eAR framework uses an edge-assisted architecture where some computational tasks are offloaded to nearby servers. This approach reduces the workload on the mobile device and helps conserve battery power while maintaining system performance [14].

Other research has explored additional strategies for improving energy efficiency in mobile AR systems. Adaptive rendering techniques can dynamically reduce the complexity of virtual objects when full detail is not required. Similarly, optimized sensor sampling methods can reduce the frequency of sensor data collection without significantly affecting system accuracy. These techniques help balance system performance with energy consumption, which is essential for mobile applications that rely on continuous environmental sensing [6].

However, many existing AR simulation systems still face challenges when handling multiple sensor inputs efficiently. Applications that rely on real-time simulations must process environmental data, update simulation parameters, and render virtual content

simultaneously. If sensor data processing is not optimized, the system may experience delays or excessive battery consumption [11].

For simulation-based AR applications such as virtual plant growth environments, efficient sensor data processing is especially important. A system that continuously monitors environmental conditions such as light levels, device movement, and user interactions must process these inputs in real time while maintaining smooth performance. Efficient sensor fusion techniques and energy-aware processing methods are therefore essential to ensure that such systems remain responsive and practical for everyday mobile use. These considerations highlight the importance of designing sensor fusion mechanisms that not only improve environmental awareness but also manage computational resources efficiently [4], [6].

Integrating optimized sensor processing with simulation engines can enable mobile AR applications to support complex interactive simulations while maintaining acceptable energy consumption levels [14].

1.2.4 Interactive Environmental Feedback in Augmented Reality Systems

Interactive feedback is a key feature of modern augmented reality (AR) systems [13]. In interactive AR environments, virtual content does not remain static; instead, it responds to user actions and changes in the surrounding environment. This interaction creates a stronger connection between the physical and virtual worlds, allowing users to feel more engaged with the AR experience. Environmental feedback mechanisms allow AR applications to monitor real-world conditions and adjust the behavior of virtual elements accordingly [8].

Many existing AR applications use user interaction and device movement as triggers for virtual events. For example, motion sensors such as accelerometers and gyroscopes are commonly used to detect gestures or device movement [5]. When a user tilts, shakes, or rotates the device, the AR system can interpret these movements as commands that trigger specific virtual effects. Some AR applications use these interactions to simulate environmental events such as wind, object movement, or physical forces. For instance, shaking a mobile device may cause virtual leaves to fall from a tree, while rotating the device can change the direction of virtual objects or camera views. These types of interactions demonstrate how sensor data can create dynamic responses inside virtual environments [8].

Environmental data can also be used to influence the behavior of virtual systems. Sensors such as ambient light detectors can provide information about the brightness of the surrounding environment [11]. When this data is used inside AR applications, virtual objects can react to environmental lighting conditions. For example, some AR systems adjust the brightness, shadows, or color of virtual objects to match the surrounding lighting environment. This helps create visual consistency between the virtual and real worlds and improves the realism of AR experiences [3].

Recent research suggests that interactive AR systems can become even more powerful when additional contextual data sources are included. Scargill et al. (2021) suggest that combining AR device sensors with data from wearable devices and Internet of Things (IoT) sensors could significantly improve environmental awareness and interaction capabilities

[2]. Wearable devices such as smartwatches or environmental sensors placed in smart environments can provide additional information about user activity and environmental conditions. When this information is integrated with AR systems, applications can respond to changes in the environment more intelligently and in real time [12].

Another important aspect of interactive AR systems is the concept of feedback loops between the real and virtual environments. In many current AR applications, interaction is mostly one-directional: the user observes virtual objects placed in the real world, but the real environment rarely influences the internal logic of the simulation [10]. For example, many AR games allow users to place virtual plants or objects in their surroundings, but the growth or behavior of those objects often depends only on preprogrammed rules rather than real environmental conditions.

Because of this limitation, the relationship between the physical environment and the virtual system is often weak. Virtual objects may appear visually integrated into the environment, but their behavior does not truly depend on real-world factors such as lighting conditions or user movement. This limits the realism and potential educational value of AR simulations [13].

To overcome this limitation, researchers have begun exploring more advanced interaction models that allow environmental conditions to directly influence virtual processes. By using environmental sensing technologies together with interactive simulation systems, AR applications can create dynamic feedback loops where real-world changes influence virtual behavior in real time [2].

For example, environmental data such as light intensity, device orientation, or user motion could influence the growth or movement of virtual elements in a simulation.

In the context of virtual plant simulations, this type of interaction can be particularly meaningful. Plants in the real world grow and respond to environmental factors such as sunlight, water availability, and physical movement caused by wind. If AR systems can capture similar environmental signals through mobile sensors, virtual plants can be designed to behave in ways that resemble natural biological processes [7].

The research proposed in this study aims to explore this type of interaction by introducing new environmental feedback mechanisms in an AR simulation environment. In the proposed system, the mobile device's flashlight will be treated as a virtual sunlight source. If the surrounding environment becomes too dark, the system can activate the flashlight and use its intensity as a parameter that supports the growth of the virtual plant. This approach allows real-world lighting conditions to directly influence the behavior of the simulated plant.

In addition, motion sensors such as the gyroscope will be used to detect device movement and simulate environmental effects such as wind or water interactions. When the user tilts or moves the device, the system can interpret this motion as environmental forces acting on the virtual plant. This creates a more natural interaction where the virtual plant responds to user movement in a realistic way.

By combining environmental sensing with interactive feedback mechanisms, the proposed system aims to create a stronger connection between the physical environment and the virtual simulation. This type of tightly coupled real-to-virtual interaction can make AR simulations more immersive, educational, and responsive to real-world conditions [2], [8].

1.3 Research Gap

Overall research problem:

How can primary learners studying the Cambridge Grade 1 Science module in Sri Lanka be better supported in understanding plant growth and environmental changes that are difficult to learn through traditional text-based teaching methods?

Individual Research Problem

How can a context-aware simulation engine be designed and developed to overcome the limitations of current AR plant simulations, which rely on static parameters, by integrating real-world environmental data such as light levels, device movement, and user interaction, in order to enable dynamic, realistic, and meaningful virtual plant growth behavior?

Although many studies have explored augmented reality technologies, procedural plant simulation, and environmental sensing methods, most of these approaches are developed separately. Existing AR systems mainly focus on improving visual realism, tracking accuracy, or interactive effects. However, limited research has investigated how real-world environmental data can be directly connected with simulation models to influence the behavior of virtual elements. Therefore, there is a need to explore new approaches that integrate environmental sensing, simulation algorithms, and interactive feedback within a single AR system.

Identified Research Gap 1 – Limited Use of Environmental Data in AR Simulations:

Existing AR systems mainly use sensor data for tracking and visual adjustments, but do not use real-world environmental data (light, motion, user interaction) to dynamically control virtual simulation behavior, resulting in static and less interactive experiences [11], [12].

Many studies have demonstrated that modern mobile devices can collect environmental data through sensors such as cameras, light sensors, accelerometers, and gyroscopes [4], [5]. Research on context-aware systems shows that this sensor data can help AR applications understand real-world conditions [2], [12]. For example, lighting estimation

techniques such as DeepLight allow AR systems to estimate environmental illumination and adjust the appearance of virtual objects accordingly [3]. Similarly, sensor fusion methods such as those proposed by Gustafsson et al. combine camera data with inertial sensor data to improve tracking accuracy and device pose estimation [4].

However, most existing AR applications use environmental sensor data mainly for improving visual alignment, object placement, and lighting effects [11]. The collected environmental information is rarely used to control or influence the internal behavior of virtual simulations. In many cases, sensor data only helps the system display virtual objects more realistically rather than making the virtual environment respond to real-world conditions.

Therefore, there is a clear research gap in using environmental sensing data as active simulation input rather than just a rendering adjustment. AR systems still lack mechanisms that translate real-world environmental conditions into parameters that directly influence the behavior of virtual objects [10], [12].

Identified Research Gap 2 – Lack of Real-Time Adaptive Plant Growth Simulation:

Current procedural plant models rely on predefined parameters and do not adapt to real-time environmental changes, preventing virtual plants from responding realistically to actual lighting conditions and user-driven interactions [18], [28].

Procedural modeling techniques have been widely used to simulate plant growth and natural ecosystems in computer graphics and virtual environments [18], [28]. Research such as the work by Alsweis and Deussen has demonstrated how biological growth processes can be modeled using computational algorithms. Their leaf growth model simulates hormone flow and tissue expansion to produce realistic plant shapes and venation patterns [28].

Other procedural approaches such as L-systems and physics-based plant models have also been widely used to generate trees, branches, and vegetation structures in virtual

environments [18]. These techniques allow complex plant forms to be generated automatically and enable virtual plants to grow over time according to defined rules.

However, most of these simulation models rely on predefined environmental parameters such as fixed light levels, water availability, or growth conditions. These parameters are usually manually defined before the simulation begins and do not change dynamically based on real-world environmental conditions.

As a result, current plant simulation systems do not reflect real environmental changes experienced by users. The growth of virtual plants often remains independent of the surrounding physical environment. This highlights an important research gap: existing procedural plant simulations rarely integrate real-time environmental sensing data to influence plant growth behavior [18], [29].

Identified Research Gap 3 – Inefficient Sensor Data Processing in Mobile AR:

Many AR systems struggle to efficiently process and combine multiple sensor inputs in real time, leading to performance issues, high battery consumption, and reduced system responsiveness during continuous simulation [6], [11].

Sensor fusion techniques have been developed to improve the accuracy and stability of AR systems [4], [5]. Combining data from cameras, inertial sensors, and environmental sensors allows AR applications to better understand device movement and environmental conditions. Research shows that sensor fusion methods can significantly improve device pose estimation and tracking reliability [4].

However, processing multiple sensor inputs continuously can create performance and energy challenges for mobile devices. Mobile AR systems require constant sensor monitoring, rendering of virtual objects, and real-time interaction processing. These operations can consume significant computational resources and reduce battery life.

Research such as the eAR framework proposed by Brocanelli highlights that mobile AR applications often face high energy consumption due to intensive rendering and sensor

processing tasks [6]. Techniques such as edge-assisted computing and adaptive rendering have been proposed to reduce computational load and improve energy efficiency [7].

Despite these improvements, many AR systems still struggle to manage sensor data efficiently when supporting complex real-time simulations. Applications that require continuous environmental sensing and simulation updates need more efficient mechanisms for aggregating and processing sensor data without affecting system performance.

Therefore, there is a research gap in designing efficient sensor aggregation and processing methods that allow AR simulation systems to handle multiple environmental inputs while maintaining smooth performance and energy efficiency on mobile devices [11], [6].

Identified Research Gap 4 – Weak Environmental Feedback Integration in AR Systems:

Existing AR applications provide limited interaction between real-world conditions and virtual behavior, lacking strong feedback loops where environmental inputs directly influence

Interactive AR systems allow virtual objects to respond to user actions and environmental conditions [8], [13]. Many AR applications already use device sensors to trigger simple interactions such as animations, object movement, or gesture-based controls. Motion sensors such as gyroscopes and accelerometers are commonly used to detect device movement and generate corresponding virtual effects [5].

Some research also suggests integrating additional context sources such as wearable devices and IoT sensors to improve interaction capabilities and environmental awareness. These approaches can enhance the responsiveness of AR systems and provide more immersive user experiences [2], [12].

However, most current AR applications still use environmental input mainly for visual effects or simple interaction triggers rather than for controlling the internal logic of simulation systems [10]. Virtual objects may appear visually integrated into the

environment, but their behavior is usually governed by predefined rules that do not respond to real-world environmental changes.

For example, many AR applications allow users to place virtual plants or objects in their surroundings, but the growth and behavior of those objects often remain unaffected by actual environmental conditions such as lighting levels or device movement.

This reveals a research gap in developing interactive AR simulation systems where real-world environmental inputs directly influence virtual processes. Current systems rarely establish a strong feedback loop between the physical environment and the internal behavior of virtual simulations [8].

Based on the above gaps, it is clear that existing research has not yet fully integrated environmental sensing, procedural plant simulation, efficient sensor processing, and interactive environmental feedback into a unified augmented reality simulation system. Most current systems treat these components separately rather than combining them to create a dynamic and context-aware virtual environment.

The proposed research addresses these gaps by developing a context-aware simulation engine that maps physical environmental data into virtual simulation parameters, allowing virtual plant growth and behavior to respond directly to real-world environmental conditions [11], [18].

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate a context-aware simulation engine for an augmented reality environment that allows a virtual plant to grow based on real-world environmental conditions.

This system aims to use environmental data collected from mobile device sensors such as light sensors, motion sensors, and user interaction inputs to influence the behavior of the virtual plant in real time. By mapping physical environmental conditions to virtual simulation parameters, the proposed system will create a more interactive and responsive AR environment.

Previous studies have shown that augmented reality systems can create engaging virtual experiences when environmental data and user interaction are effectively integrated into the system design. However, many existing AR applications focus mainly on visual rendering, object placement, or basic interaction effects rather than connecting real-world environmental conditions with simulation behavior.

Research on procedural plant simulation also demonstrates that virtual plants can be generated using mathematical growth models and procedural algorithms. However, these models often rely on predefined environmental parameters instead of responding to real-time environmental data collected from the user's surroundings.

Similarly, research on sensor fusion in mobile AR systems highlights that combining data from multiple sensors such as cameras, accelerometers, and gyroscopes can improve environmental awareness and tracking accuracy. Despite these advancements, many AR simulation systems still lack efficient mechanisms to process environmental sensor data and translate it into simulation behaviors.

Therefore, this research aims to bridge these limitations by developing a context-aware simulation framework that integrates environmental sensing, procedural plant growth

modeling, and interactive feedback mechanisms within a unified AR system. The system will allow virtual plant growth to dynamically respond to environmental conditions such as light intensity, device movement, and user interaction, creating a more realistic and interactive virtual ecosystem.

2.2 Specific Objectives

To achieve the main objective, the study will pursue the following specific objectives. These objectives divide the research into several clear and manageable components so that the system can be designed, implemented, and evaluated in a systematic way.

The first objective focuses on **environmental sensing**, where the system will detect lighting conditions in the surrounding environment using mobile device sensors. This allows the application to understand whether the environment is bright or dark and adjust the simulation accordingly.

The second objective focuses on **procedural plant growth simulation**. In this stage, an algorithm will be designed to simulate the growth behavior of a virtual plant based on environmental conditions such as light exposure and user interaction. This will allow the plant to grow dynamically instead of following a fixed predefined pattern.

The third objective focuses on **sensor fusion and efficient data processing**. Since the system will use multiple sensors simultaneously, an efficient mechanism will be implemented to aggregate and process sensor data while maintaining system performance and minimizing battery consumption.

The fourth objective focuses on **interactive environmental feedback mechanisms**. In this stage, additional interaction methods will be introduced to strengthen the connection between the real and virtual environments. For example, the device flashlight can act as a virtual sunlight source to support plant growth when the surrounding environment is too

dark. Similarly, gyroscope data can be used to simulate environmental effects such as wind or water movement in the virtual environment.

By achieving these objectives, the proposed system aims to create a more responsive and context-aware AR simulation environment where virtual plant growth is influenced by real-world environmental conditions.

Table2.1: Specific Objectives

Objective No	**Specific Objective**
SO1	Develop a method to detect environmental light levels using mobile device sensors so that the system can understand whether the surrounding environment is bright or dark.
SO2	Design an algorithm that simulates virtual plant growth based on environmental conditions such as light exposure, user interaction, and simulated environmental factors.
SO3	Implement an efficient sensor data processing mechanism that combines data from multiple sensors while maintaining smooth performance and minimizing battery consumption.
SO4	Develop an interactive feature that uses the mobile device flashlight and motion sensors to simulate environmental effects such as sunlight and wind within the virtual plant growth simulation

3. METHODOLOGY

The research methodology includes **prototype development, controlled experimental testing, and performance evaluation**. During system testing, environmental sensor data, simulation behavior, and system performance will be recorded to analyze how effectively the virtual plant simulation responds to real-world environmental conditions. The evaluation process will focus on system responsiveness, simulation realism, sensor accuracy, and overall performance efficiency.

Since the research involves interaction with a mobile application rather than sensitive personal data collection, the study will primarily collect **environmental sensor readings, system logs, and simulation response data**. No personal or identifiable user information will be stored. All data collected during testing will be used solely for research purposes and will follow basic ethical research practices such as anonymization and responsible data handling.

The overall methodology is organized according to the four key research components identified in the literature review and research objectives, namely context-aware environmental sensing, procedural plant growth simulation, efficient sensor data processing, and interactive environmental feedback mechanisms**.**

Each component represents a specific objective of the research and contributes to the development of the proposed context-aware AR simulation framework.

Table 3.1: Methodology for Context-Aware Environmental Sensing in Mobile AR

Aspect	**Methodology**
Research approach	This component follows a prototype-driven design approach where environmental sensing capabilities are implemented and tested within a mobile AR application. The aim is to detect real-world environmental conditions such as lighting and device motion and use this data to influence the AR simulation environment.
Technical methods and algorithms	Environmental sensing will be implemented using mobile device sensors including ambient light detection, camera-based illumination estimation, and motion sensing through gyroscope and accelerometer data. Sensor fusion techniques will be applied to combine multiple sensor inputs and produce stable environmental parameters that can be used by the simulation engine.
Tools and platforms with justification	Unity 3D and AR Foundation will be used for AR application development because they provide built-in support for sensor integration, camera-based environmental analysis, and cross-platform mobile AR deployment. These tools allow real-time access to mobile device sensors required for environmental sensing.
Data collection and ethical considerations	Environmental data such as light intensity and device motion will be collected through mobile device sensors during system testing. No personal user information will be collected. All sensor data will be used only for system testing and performance evaluation purposes.
Validation strategy and performance metrics	Validation will focus on the accuracy and responsiveness of environmental sensing. Performance metrics include light detection accuracy, sensor stability, and the system's ability to respond to changes in environmental conditions in real time.

Table 2.2: Methodology for Procedural Plant Growth Simulation

Aspect	**Methodology**
Research approach	This component follows a simulation-based development approach where a procedural plant growth model is designed and integrated into the AR environment. The aim is to simulate plant growth behavior that dynamically responds to environmental parameters detected by the system.
Technical methods and algorithms	A procedural growth algorithm will be developed to simulate plant development over time. The algorithm will define growth rules that depend on environmental inputs such as light intensity and user interaction. Growth parameters such as plant height, leaf expansion, and growth rate will change dynamically according to environmental conditions.
Tools and platforms with justification	Unity 3D will be used to implement the simulation logic and visualize plant growth in real time. The platform supports procedural modeling and real-time rendering, which are necessary for implementing dynamic plant growth simulations within an AR environment.
Data collection and ethical considerations	Data collected during testing will include simulation parameters such as plant growth rate, environmental input values, and system response behavior. No personal data from users will be recorded. All testing will be conducted in controlled environments.
Validation strategy and performance metrics	Validation will focus on the realism and responsiveness of the plant growth simulation. Performance metrics include growth response accuracy, simulation stability, and consistency of plant behavior under different environmental conditions.

Table 3.3: Methodology for Sensor Fusion and Energy-Efficient Data Processing

Aspect	**Methodology**
Research approach	This component follows an experimental development approach where sensor data processing methods are implemented and optimized to ensure efficient system performance. The aim is to combine multiple sensor inputs while maintaining stable performance and minimizing device resource usage.
Technical methods and algorithms	Sensor fusion techniques will combine gyroscope, accelerometer, and environmental sensor data to generate stable simulation parameters. Efficient data aggregation methods will be used to reduce unnecessary sensor processing and maintain real-time system responsiveness.
Tools and platforms with justification	Unity's mobile optimization tools and device sensor APIs will be used to manage sensor data processing. These tools allow efficient handling of sensor streams while maintaining stable frame rates and minimizing energy consumption.
Data collection and ethical considerations	System performance data such as processing time, sensor update frequency, and battery usage will be recorded during testing. No personal user data will be collected during the research process.
Validation strategy and performance metrics	Validation will evaluate system performance under continuous sensor processing. Metrics include processing efficiency, system responsiveness, frame rate stability, and battery consumption during application usage.

Table 3.4: Methodology for Interactive Environmental Feedback in AR Systems

Aspect	**Methodology**
Research approach	This component follows an interaction design and evaluation approach where environmental feedback mechanisms are implemented and tested within the AR simulation. The goal is to create a stronger interaction between the real environment and virtual plant simulation.
Technical methods and algorithms	Interactive mechanisms will be developed where the device flashlight acts as a virtual sunlight source and gyroscope data simulates environmental forces such as wind. The system will map these environmental inputs into simulation parameters that influence plant growth and behavior.
Tools and platforms with justification	Unity 3D and AR Foundation will be used to implement real-time interaction mechanisms. These platforms support device hardware access such as flashlight control and motion sensor input required for environmental interaction.
Data collection and ethical considerations	Data collected during testing will include interaction logs, environmental sensor readings, and system response to user input. Testing will not involve collecting personal or sensitive information.
Validation strategy and performance metrics	Validation will focus on interaction responsiveness and realism. Metrics include system response time, accuracy of environmental interaction effects, and user feedback on the realism of plant-environment interaction.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed solution is a handheld Augmented Reality (AR) storybook system designed to support Cambridge Primary Science Grade 1 learning in Sri Lankan international schools. The system combines a printed storybook with mobile AR visualizations, interactive learning activities, and child-friendly feedback mechanisms to create an engaging and developmentally appropriate learning experience. When a learner points a mobile device at a storybook page, the system detects the page, overlays relevant virtual content, and enables simple guided interaction through touch and visual prompts. The overall solution is intended to support both learning engagement and concept understanding while maintaining usability, comfort, and age-appropriate interaction. At the system level, the proposed solution consists of four main components contributed by the four group members. These components work together as an integrated learning system. The first component focuses on content delivery and educational interaction, which manages the story flow, lesson progression and presentation of science learning material. The second component focuses on assessment and learner response handling, which captures learner actions and supports activity-based evaluation. The third component focuses on visual AR rendering and object presentation, which controls how digital learning content is displayed on top of the physical storybook. The fourth component, which is the focus of this study, is the hybrid interaction framework, which manages child-friendly mobile interaction, occlusion handling and anchoring stability, handheld ergonomic flow, and cognitive load reduction. Together, these components form a complete AR learning environment.

4.2 System Diagram of Components

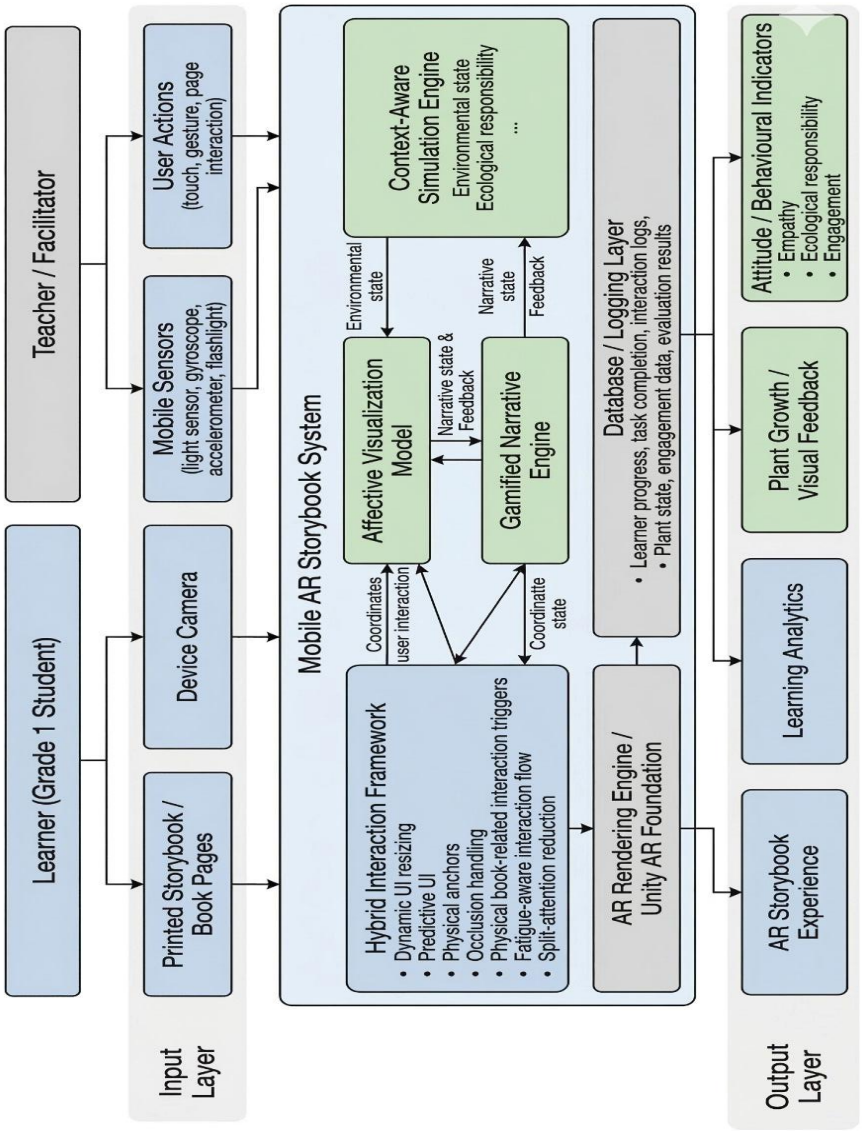


Figure 1:System Diagram of Components

4.3 Individual Component and Its Parts

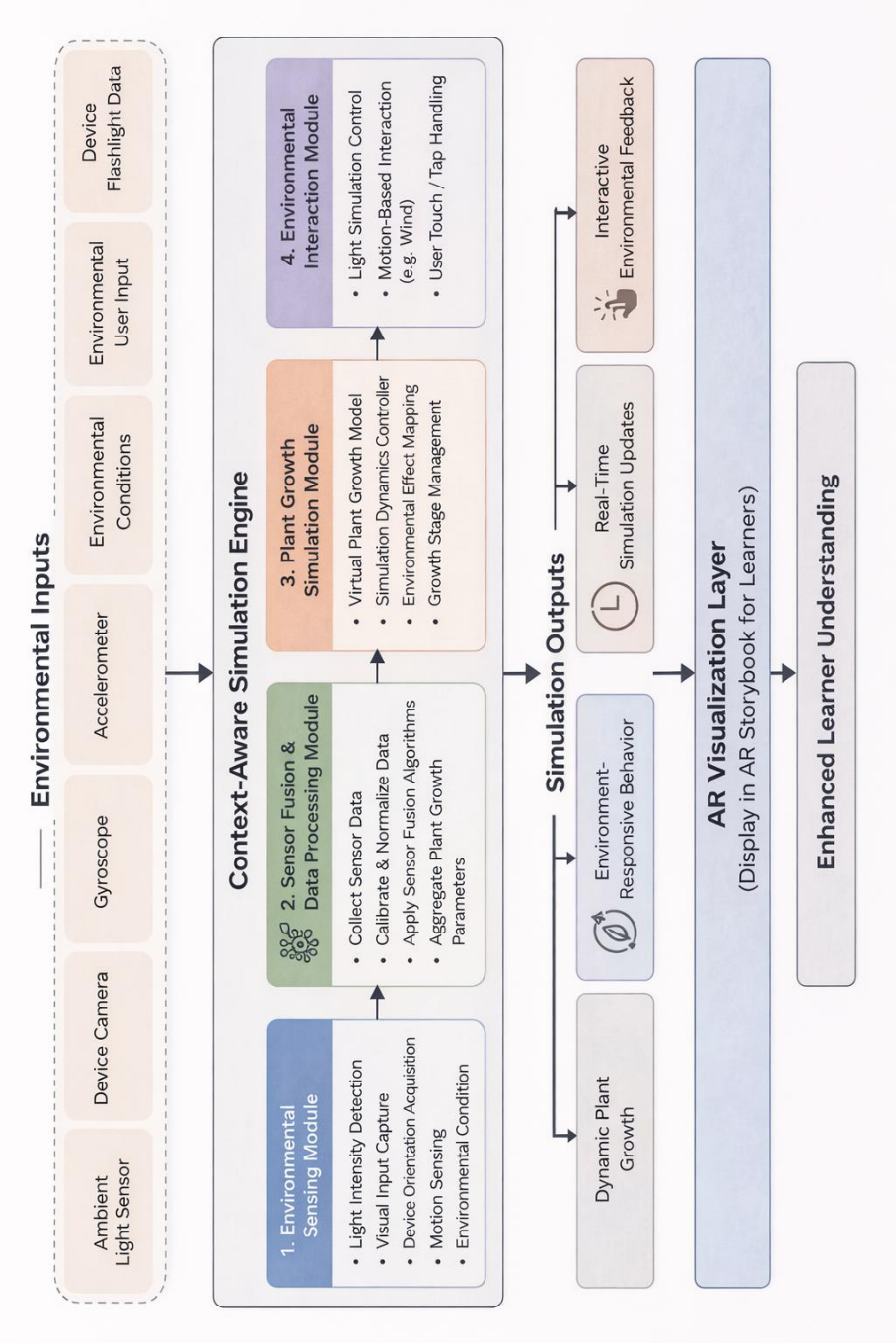


Figure 2:Individual Component and Its Part

4.4 Component Interactions

The proposed system operates through interactions between several modules that work together to simulate virtual plant growth based on environmental conditions. The process begins with the **Environmental Input**, where real-world conditions such as light and device movement are detected through mobile sensors.

The **Environmental Sensing Module** collects raw data from sensors including the camera, light sensor, gyroscope, and accelerometer. This data is then processed in the **Sensor Processing and Fusion Module**, where multiple sensor inputs are combined to generate stable environmental parameters.

These parameters are sent to the **Context-Aware Simulation Engine**, which controls the growth and behavior of the virtual plant according to environmental conditions. The **Interactive Feedback Module** enables additional interactions such as using the device flashlight as virtual sunlight and using motion sensors to simulate wind effects.

Finally, the **AR Visualization Module** displays the virtual plant within the real-world environment through the mobile AR interface, allowing users to observe how the plant responds to environmental changes in real time.

4.5 Data Flow

The data flow of the proposed system describes how environmental information is captured, processed, and used to control the virtual plant simulation within the AR environment.

First, environmental data such as light intensity and device motion are captured through the mobile device sensors including the camera, light sensor, gyroscope, and accelerometer. These raw sensor readings represent the real-world environmental conditions around the user.

Next, the collected sensor data is sent to the Sensor Processing and Fusion Module, where the data is cleaned, combined, and converted into meaningful environmental parameters. This step ensures that the system produces stable and accurate inputs for the simulation.

The processed environmental parameters are then passed to the Context-Aware Simulation Engine. In this stage, the system maps the environmental inputs to simulation rules that control the behavior of the virtual plant. For example, higher light intensity may increase plant growth, while device movement may simulate environmental effects such as wind.

After the simulation engine updates the plant state, the updated information is forwarded to the Interactive Feedback Module. This module allows additional environmental interactions such as activating the device flashlight to simulate sunlight or using motion sensors to create environmental effects.

Finally, the updated plant model and simulation results are sent to the AR Visualization Module, which renders the virtual plant within the real-world environment using the mobile camera feed. The user can observe how the plant grows and reacts dynamically to environmental conditions and interactions.

Through this continuous flow of data between sensors, processing modules, simulation components, and visualization layers, the system creates a responsive and context-aware AR simulation environment.

4.6 Scalability and Security Considerations

The proposed system is designed with scalability and security considerations to ensure that it can operate efficiently across different mobile devices and maintain safe handling of data. Scalability is important because the system may be extended in the future to support additional environmental sensors, more complex plant simulation models, or multiple virtual objects within the AR environment.

From a scalability perspective, the system uses a modular architecture where each component, such as environmental sensing, sensor processing, simulation, and AR visualization, operates independently but communicates through defined data flows. This modular design allows new sensors or simulation features to be integrated without significantly modifying the entire system. Additionally, the use of mobile AR frameworks such as Unity AR Foundation supports deployment across different platforms, enabling the system to scale across various mobile devices and operating systems.

Security considerations are also addressed to ensure that the system handles data responsibly. The application primarily collects environmental sensor data such as light intensity and device motion, which does not include personal or sensitive user information. All sensor readings and system logs collected during testing will be used only for research and system performance evaluation.

Access to device hardware components such as the camera, sensors, and flashlight will follow the permission mechanisms provided by the mobile operating system, ensuring that users are informed about sensor usage. Any stored experimental data will be anonymized and protected from unauthorized access.By incorporating these scalability and security measures, the proposed system ensures that the AR simulation framework can be safely expanded in the future while maintaining responsible handling of system data and device resources.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholder of the proposed system is the **Grade 1 learner**, because the context-aware AR plant simulation is designed for children to interact directly with the virtual plant environment using a mobile device. The learner interacts with the system by observing the AR environment, moving the device, adjusting lighting conditions, and interacting with the virtual plant. Therefore, the system must be designed to match the child's basic interaction abilities, short attention span, and simple understanding of environmental concepts.

The **teacher or classroom facilitator** is another important stakeholder. Teachers guide students during classroom activities and help them understand how environmental factors such as light and movement affect plant growth in the AR simulation. The system should therefore provide clear visual feedback and simple interactions so that teachers can easily demonstrate concepts and support student learning.

Parents are also considered supporting stakeholders because they are concerned about the safety, usability, and educational value of digital learning applications used by young children. The system should therefore be simple, safe, and easy for children to use with minimal supervision.

Finally, **school administrators and educational institutions** are supporting stakeholders because they are responsible for adopting and approving educational technologies used in classrooms. They are concerned with how well the system supports learning, how easily it can be integrated into classroom activities, and whether it provides educational value for young learners.

These stakeholders align with the proposed AR plant simulation system, which combines **mobile sensors, environmental interaction, and virtual plant growth visualization** to help young learners understand basic environmental concepts in an engaging and interactive way.

5.2 Functional Requirements

The context-aware AR plant simulation system shall support the following functional requirements:

FR1. The system shall detect real-world environmental conditions such as light intensity using mobile device sensors and use this information as input for the virtual plant simulation. This requirement allows the system to understand the environmental context surrounding the user.

FR2. The system shall simulate virtual plant growth dynamically within the augmented reality environment based on environmental parameters such as light exposure and user interaction. This requirement ensures that the virtual plant behaves realistically according to environmental conditions.

FR3. The system shall combine data from multiple sensors such as the camera, light sensor, gyroscope, and accelerometer in order to generate stable environmental parameters for the simulation engine. This supports reliable environmental sensing and system stability.

FR4. The system shall provide interactive environmental feedback mech**anisms** where device movement and environmental changes influence the behavior of the virtual plant. For example, device motion may simulate environmental effects such as wind.

FR5. The system shall allow the mobile device flashlight to act as a virtual sunlight **source** when the surrounding environment is too dark, supporting the growth of the virtual plant.

FR6. The system shall display the virtual plant within the real-world environment using augmented reality visualization, allowing users to observe plant growth and environmental interactions in real time.

FR7. The system shall continuously update the simulation in real time as environmental conditions change, ensuring that the virtual plant responds dynamically to user actions and surrounding environmental inputs.

5.3 Non-Functional Requirements

The non-functional requirements are important because they ensure that the proposed context-aware AR plant simulation system operates efficiently, safely, and effectively for young learners.

NFR1. Usability:

The system must be simple and easy to use for Grade 1 learners. The interface should present clear visuals and simple interactions so that children can easily observe and understand the virtual plant growth simulation. Since the system relies on environmental interaction such as lighting and device movement, usability ensures that students can interact with the AR environment without confusion or complex controls.

NFR2. Performance:

The system must process environmental sensor data and update the virtual plant simulation in real time. The AR application should maintain smooth performance and stable visualization so that the plant growth and environmental effects respond quickly to changes in lighting conditions or device movement.

NFR3. Reliability:

The system must consistently detect environmental conditions and maintain stable AR tracking during operation. Sensor data processing and simulation updates should function reliably so that the virtual plant remains correctly positioned and behaves consistently during user interaction.

NFR4. Safety:

The system must ensure safe usage for young learners. Since the application uses device sensors and flashlight features, it should operate within safe limits and avoid excessive brightness or device overheating. The system should also avoid collecting personal or sensitive data from users.

NFR5. Energy Efficiency:

The system should minimize battery consumption while processing environmental sensor data and running the AR simulation. Efficient sensor data processing and optimized rendering should be implemented so that the application can run for longer periods without significantly draining the mobile device battery.

NFR6. Compatibility:

The system should function across commonly used mobile devices that support augmented reality features. Using platforms such as Unity AR Foundation allows the application to operate across different mobile operating systems and devices.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed solution is international schools in Sri Lanka that follow the Cambridge Primary Science curriculum, especially schools that seek interactive and technology-supported learning tools for early primary education. The system is designed as a handheld AR storybook that runs on standard ARCore- and ARKit-enabled smartphones and tablets, which makes it suitable for schools that already use mobile devices or tablets in classroom activities. Although the AR storybook is developed according to the Cambridge Grade 1 Science syllabus in international schools, it is also appropriate for 6-year-old children who enjoy learning about plants, because the concept, visuals, and interaction style are intended for early learners.

Table 3.1: Primary Customer Persona

Category	Details
Persona Name	School Decision-Maker
Role	Principal / Sectional Head / ICT Coordinator / Primary Curriculum Coordinator
Type	Primary Customer
Profile	Responsible for selecting and approving educational technology solutions for the school.
Needs	A reliable, curriculum-aligned, safe, and innovative learning tool that improves student engagement and supports classroom teaching.
Goals	To introduce effective digital learning solutions that enhance educational quality and support modern teaching methods.
Pain Points	Difficulty finding age-appropriate AR tools that are educationally meaningful, easy to implement, and suitable for young learners.
How the System Helps	The AR storybook provides a research-based, child-friendly, and curriculum-relevant learning solution for Grade 1 science education.

Table 6.2: Secondary Customer Persona

Category	Details
Persona Name	Grade 1 Science Teacher
Role	Early Primary Science Teacher
Type	Secondary Customer / Direct User
Profile	Uses the system in classroom teaching to help students understand plant-related science concepts in an interactive way.
Needs	A classroom-friendly tool that is easy to use, engaging for children, and supportive of concept understanding.
Goals	To make science learning more interesting, understandable, and memorable for Grade 1 learners.
Pain Points	Traditional teaching methods may not always keep young learners engaged or help them clearly visualize plant-related concepts.
How the System Helps	The AR storybook combines storytelling, visuals, and guided interaction to make science lessons more engaging and easier to understand.

Table 6.3: Indirect Customer Persona

Category	Details
Persona Name	Parent of a Grade 1 Learner
Role	Parent / Guardian
Type	Indirect Customer
Profile	Supports the child's learning at home and values meaningful educational experiences beyond the classroom.
Needs	A safe, engaging, and educational tool that helps the child learn effectively in a guided or supervised setting.
Goals	To support the child's curiosity, enjoyment, and understanding of science concepts at home.
Pain Points	Limited access to interactive educational tools that are both enjoyable and academically useful for young children.
How the System Helps	The AR storybook supports emotional engagement, guided learning, and progress-related feedback, making it useful for supervised home learning as well.

6.2 Value Proposition

The main value proposition of the proposed system is that it offers a child-centered, safe, and engaging AR learning experience that combines physical storybooks with interactive digital content to improve understanding of plant-related science concepts. The proposed solution already emphasizes that printed storybook pages, camera input, mobile sensor data, and guided interactions are integrated to create an AR learning experience that is more meaningful than static textbook diagrams. In addition, the system is differentiated by its coordinated research modules, particularly the Hybrid Interaction Framework, which addresses touch accessibility, anchoring between physical pages and 3D content, split-attention reduction, occlusion handling, and fatigue-aware interaction support.

From a commercialization perspective, this gives the product clear value for schools: it is not just an AR storybook, but a research-driven educational tool designed specifically for young learners' usability, comfort, and engagement needs. This makes it more attractive than generic AR applications that may focus mainly on visual novelty without considering children's fine motor skills, physical comfort, or cognitive load.

6.3 Revenue Model

A suitable revenue model for this product is a business-to-school licensing model. Under this model, schools would pay an annual or term-based subscription for access to the AR storybook application, learning content, teacher support features, and software updates. This approach is practical because the system is intended for structured educational use rather than one-time casual entertainment. A second possible model is a per-classroom or per-device licensing structure, where schools pay based on the number of student devices or class groups using the system.

An additional revenue stream could come from content expansion packs, such as new storybooks, additional science topics, or curriculum-aligned activity sets for later grades. Since the proposed architecture is modular and includes multiple coordinated research components, the solution has the potential to be extended beyond the initial plant-growth topic into broader environmental or science learning domains. In the longer term, the product could also be offered through partnerships with educational publishers or private tuition and learning centers.

6.4 Cost Estimation

The main cost categories of commercialization would include software development and refinement, content production, 3D asset and animation development, testing and quality assurance, teacher onboarding materials, and maintenance/support costs. The project requires practical development with Unity 3D (2024 LTS), AR Foundation, ARCore, and ARKit, as well as ongoing prototyping, iteration, and testing. These factors suggest that the early commercialization stage would involve costs related to continued technical development, optimization, and device compatibility testing.

There would also be costs for producing high-quality storybook pages, visual assets, and classroom-ready deployment materials. If the product is marketed to schools, teacher guides, setup instructions, and user support documentation would also be necessary. In the pilot stage, costs may be relatively high per user because the product is still being refined. However, once the core application and content pipeline are established, the marginal cost of distributing the software to additional schools would be lower than that of physical-only learning tools.

6.5 Pricing Strategy

The pricing strategy should be based on institutional affordability and educational value. A practical approach would be to offer a school subscription package with different tiers depending on the number of users, classes, or content modules required.

Because the target customers are schools rather than individual mass-market consumers, the pricing should remain reasonable enough for adoption while also reflecting the research-based and curriculum-aligned value of the product. A pilot pricing strategy may also be used in the initial stage, where selected schools receive the system at a discounted rate in exchange for implementation feedback and case-study data. This would support both early adoption and product refinement.

6.6 Competitive Advantage

The proposed solution has several competitive advantages. First, it combines physical storytelling with handheld AR in a way that is explicitly designed for young learners rather than adapted from adult-centered AR systems. The Hybrid Interaction Framework includes predictive UI, dynamic button resizing, physical book-related interaction triggers, and fatigue-related interaction design to improve usability and comfort for children in handheld AR learning. This gives the product a strong advantage over many existing AR learning tools that focus mainly on visual interaction without addressing motor accessibility, ergonomic comfort, or split-attention reduction.

Second, the product is built as an integrated educational ecosystem rather than as a standalone AR animation tool. The complete system includes a Hybrid Interaction Framework, an Affective Visualization Model, a Narrative Engine, and a ContextAware Simulation Engine working together as coordinated modules. This means the product can offer emotionally responsive visuals, story-driven engagement, and realworld sensor-based interaction in addition to usability improvements. Third, because the solution is intended for early science education and is aligned to a school curriculum context, it has stronger classroom relevance than general-purpose AR entertainment applications.

6.7 Intellectual Property Considerations

The intellectual property (IP) of the proposed product would likely include several protectable elements: the software implementation, the interaction design logic, the storybook content, the 3D assets and animations, and possibly the novel integration of hybrid interaction mechanisms such as adaptive UI resizing, physical anchors, book related triggers, and fatigue-aware flow design. These aspects may support future claims for software ownership, design protection, or publication-based research recognition.

At the commercialization stage, it will be important to clearly determine the ownership of the software, assets, and content among the student researchers, the university, and any future industry partners. Copyright protection would apply to the storybook design, visuals, animations, and source code, while trademark protection may later be considered for the product name and brand identity. If the final implementation contains sufficiently novel technical methods, there may also be potential for patent-related discussion, although that would require legal and institutional review. In addition, care must be taken to avoid infringement when using third-party libraries, asset packs, or audio resources during product development.

7. BUDGET AND JUSTIFICATION

7.1. Hardware Costs

The development and testing of the context-aware simulation component require hardware devices capable of capturing real-time environmental data and supporting mobile augmented reality processing. Since this component depends heavily on environmental sensing and real-time simulation, the selected hardware must include reliable sensors such as cameras, ambient light sensors, gyroscopes, and accelerometers.

A mid-range Android smartphone with ARCore support is required as the primary device for implementing and testing the simulation engine. This device enables real-time data collection from sensors, including light intensity detection for plant growth simulation and motion tracking for environmental interaction such as wind effects. Additionally, the device flashlight will be used as a virtual grow light within the simulation.

A stable testing setup is also required, including a simple mobile holder or stand, to conduct controlled experiments when evaluating sensor-based interactions and simulation behavior. Printed storybook pages or markers will be used alongside the device to test AR tracking and integration with the simulation system.

These hardware components are essential to ensure accurate environmental sensing, smooth simulation processing, and reliable interaction between real-world inputs and virtual plant behavior within the AR storybook system.

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Table 7.1: Hardware Cost Estimation

Category	Description	Estimated Cost (LKR)
Mobile Device	ARCore supported Android smartphone (for simulation + sensor input)	35,000
Printed Materials	Storybook pages / AR markers for simulation testing	5,000
Device Accessories	Mobile stand / holder for controlled testing	3,000
Total Estimated Cost		**43,000**

7.2. Software Licenses

To develop the context-aware simulation engine, several software tools will be required for implementing environmental sensing, simulation logic, and AR integration. Most of these tools provide free or student versions, which helps reduce development costs.

The Unity 3D platform will be used as the main development environment to implement the simulation engine and integrate real-time environmental data into virtual plant behavior. AR Foundation, along with ARCore and ARKit, will be used to enable mobile augmented reality functionality and sensor-based interaction.

Programming and simulation logic will be implemented using C# within Unity, while Visual Studio or Visual Studio Code will be used for coding, debugging, and performance optimization. Sensor data processing and integration (such as light sensor, gyroscope, and accelerometer inputs) will be handled using Unity's built-in APIs.

Additionally, tools such as Blender or Autodesk Maya may be used if needed to create or modify 3D plant assets required for testing simulation behavior under different environmental conditions. Visual effects related to environmental interaction (such as lighting and motion effects) will be implemented using Unity tools like Shader Graph and Particle System.

Since most of these tools are available as free versions or student licenses, the overall software cost remains minimal while still supporting full implementation of the simulation component.

Table 7.2: Software License Cost Estimation

Category	Description	Estimated Cost (LKR)
Unity 3D	Main development platform (Free Personal License)	0
AR Foundation + ARCore / ARKit	AR development and sensor integration	0
Visual Studio / VS Code	Programming and debugging tools	0
Blender / Autodesk Maya	3D model creation (student/free versions)	0
Unity Tools (Shader Graph, Particle System)	Visual effects and simulation support	0
Total Estimated Cost		**0**

7.3. Cloud Services

Cloud services may be used to support optional features related to the context-aware simulation component, such as storing environmental data logs, simulation performance data, and testing results. Although the core simulation engine operates locally on the mobile device using real-time sensor data, cloud services can assist in managing and analyzing collected data during evaluation phases.

During testing, sensor data such as light intensity, device motion, and interaction logs can be stored in cloud platforms to analyze system performance, simulation accuracy, and user behavior. This allows developers to evaluate how effectively the simulation responds to real-world environmental conditions.

Cloud services can also support updates to simulation parameters, enabling improvements to plant growth models and environmental interaction logic without requiring major application changes. Additionally, cloud storage can be used to maintain backups of testing data and system configurations.

Basic cloud platforms such as Google Drive, Firebase, or similar free-tier services can be used for data storage and synchronization. Since the simulation primarily runs on-device and cloud usage is limited to optional data storage and analysis, the overall cloud cost remains minimal while still supporting scalability and evaluation needs.

Table 7.3 : Cloud Services Cost Estimation

Category	Description	Estimated Cost (LKR)
Cloud Storage	Store sensor data logs, simulation results, testing data (Google Drive / Firebase free tier)	0
Data Backup	Backup of simulation configurations and evaluation data	0
Content Updates	Update simulation parameters and system improvements	0
Total Estimated Cost		**0**

7.4. Human Effort Cost

Human effort for the context-aware simulation component involves the time and work required to design, implement, test, and optimize the simulation engine that integrates real-world environmental data into virtual plant behavior. This includes developing simulation algorithms, integrating sensor data, optimizing performance, and conducting testing and evaluation.

Since this component heavily relies on environmental sensing and real-time processing, significant effort is required for implementing sensor data handling, simulation logic, and ensuring efficient performance on mobile devices. Additional effort is also needed for testing how accurately the simulation responds to environmental conditions such as light and device movement.

As the development is carried out by undergraduate students, the estimated cost reflects a reasonable academic-level valuation rather than professional industry rates. Minor additional costs may include travel for school visits, testing sessions with children, and preparation of evaluation materials.

Category	Description	Estimated Cost (LKR)
Development Effort	Simulation engine design, sensor integration, algorithm implementation	10,000
Testing & Evaluation	Conducting experiments, analyzing simulation accuracy, user testing	8,000
Travel & Miscellaneous	School visits, testing sessions, printing materials	5,000
Total Estimated Cost		**23,000**

Table 7.4: Total Estimated Budget

Category	Description	Estimated Cost (LKR)
Hardware Costs	Mobile device, printed materials, accessories for simulation testing	43,000
Software Licenses	Unity, AR tools, programming tools, 3D modeling tools (free versions)	0
Cloud Services	Data storage, simulation logs, updates (free-tier services)	0
Human Effort Cost	Development, testing, evaluation, and travel	23,000
Total Estimated Budget		**66,000 LKR**

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) of the proposed research is organized according to the four integrated components of the handheld AR storybook system. In the proposed solution, the system operates through printed storybook pages, camera and sensor inputs, and user actions, while the process layer is divided into four coordinated modules: the Hybrid Interaction Framework, the Affective Visualization Model, the Gamified Narrative Engine, and the Context-Aware Simulation Engine. The final outputs are expected to include the AR storybook experience, learning analytics, and attitude-related indicators. This modular structure provides a clear basis for dividing work among the team members and supports systematic development and integration of the overall product.

Table 8.1: Task Breakdown per Member

Member	Component	Task Breakdown
IT22641410 – Wijayarathne K.N.	Hybrid Interaction Framework	Design and implement the dynamic UI mechanism for children's fine motor support; develop physical anchors and occlusion handling for printed storybook interaction; design fatigue-aware interaction flow to reduce prolonged arm strain; conduct usability testing related to cognitive load, engagement, and splitattention reduction
IT22179258 – Palliyaguruge S.N.	Affective Visualization Model	Design AR-optimized 3D plant assets; create animations for plant health, stress, and recovery; apply visual and audio effects to improve emotional readability; conduct child-centered AR trials to refine interpretation and empathy-oriented visual performance.

IT22237804 – Amarawardhane A.S.	Gamified Narrative Engine	Develop the branching story structure; create AI/rule-based dialogue and feedback according to plant condition and context; implement time-delta persistence for plant growth and recovery; design redemption-based rewards and evaluate environmental attitudes and responsibility.
IT22626974 – Dissanayaka D.M.D.D.	Context-Aware Simulation Engine	Develop light-sensing functionality; create algorithms to simulate plant growth based on environmental conditions and user actions; optimize sensor aggregation for efficiency; implement flashlight-assisted growth support and gyroscope-based wind/water simulation.
All Members	Integration and Evaluation	Integrate the four modules into one handheld AR storybook system; perform joint system testing, debugging, user evaluation, refinement, presentation preparation, and final product completion

Table 8.2: Timeline Alignment

Time Period	Planned Activities	Related Deadline
March 2026	Finalize proposal, refine requirements, confirm component scope, and prepare initial designs for all four modules.	Proposal Submission / Proposal Presentation
April 2026	Begin early prototype development for interaction design, visualization, narrative logic, and simulation logic.	Proposal milestone continuation
May 2026	Complete first-stage implementation of core module functions and prepare for initial review.	Progress Presentation 1 / Check List Submission
June 2026	Continue implementation and begin cross-component integration, including AR page flow, module communication, and output handling.	Post-Progress 1 development phase

July 2026	Conduct usability testing, debugging, integration refinement, and preliminary evaluation of the overall prototype.	Preparation for Progress Presentation 2
August 2026	Complete the full product, finalize working prototype, and ensure all four	Progress Presentation 2
	modules function together as one integrated system	
September 2026	Improve stability, finalize evaluation, organize documentation, and prepare report and website content.	Final documentation phase
October 2026	Submit final report, website, final checklist, and complete final presentation and viva.	Final submission milestones
December 2026	Prepare and submit research paper publication evidence	Research paper publication evidence submission

Table 8.3: Balanced Workload Distribution

Work Area	Workload Distribution Explanation
Component Ownership	Each member is assigned one major research component, which creates clear responsibility and avoids unnecessary overlap. This supports balanced distribution across interaction, visualization, narrative, and simulation development.
Technical Development	The work is balanced because all four members develop a comparable subsystem with its own design, implementation, and testing responsibilities.

Integration Work	Integration is shared because the final system depends on interaction among all four modules inside one AR storybook platform. This includes combining sensor input, page tracking, story flow, visualization, and usability support.
Testing and Evaluation	Evaluation is also shared, since the project collects observation data, interaction logs, plant growth status, and pre/post outcomes, and requires ethical handling of child participants.
Final Deliverables	Report writing, presentation preparation, website completion, and final refinement are group responsibilities, ensuring that no single member carries all final-stage work.

9. GANTT CHART

Table 9.1: GANTT CHART

Research Activities	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**
Literature Review & Problem Analysis	■							
Research Proposal Preparation	■							
System Architecture Design	■	■						
Environmental Sensing Module Development		■	■					
Plant Growth Simulation Development		■	■					
Sensor Data Processing & Optimization		■	■					
Interactive Environmental Feedback Implementation		■	■					
System Integration & Prototype Development		■	■					
Progress presentation 1			■					
System Testing & preparation				■	■			
Component level testing and debugging				■	■	■		
Evaluation of usability, comfort, and cognitive load				■	■	■		
Final integration and refinement				■	■	■	■	
Progress presentation 2						■		
Final Report Writing & Documentation						■	■	■
Final deliverable preparation							■	■
Final Project Submission & Viva								■

Legend = ■ active period

This Gantt chart shows that the context-aware AR plant simulation research begins with literature review, requirement analysis, and system design planning during the initial proposal stage. The development phase then focuses on four key technical components: environmental sensing, procedural plant growth simulation, sensor data processing and fusion, and interactive environmental feedback mechanisms.

The middle phase of the schedule is allocated for system implementation, module integration, and prototype testing to ensure that the simulation engine correctly responds to environmental conditions. The later phase of the timeline focuses on system evaluation, performance analysis, documentation, and preparation of the final research report.

This timeline ensures that the proposed context-aware AR simulation system is developed, tested, and refined within the project schedule and is ready for demonstration and final evaluation before the final research presentation.

10. REFERENCES LIST

[1] A. Eka Lidiastuti, A. Arinda, and others, “GrowLab: The Effect of Augmented Reality-Based Learning on Plant Science Education,” *Edubiotik Journal,* pp. 356-364.

[2] T. Scargill, Y. Chen, S. Eom, J. Dunn, and M. Gorlatova, “Environmental, User, and Social Context-Aware Augmented Reality for Supporting Personal Development and Change,” in *Proc. IEEE PerCom Workshops,* 2021.

[3] P. Kán and H. Kaufmann, “DeepLight: Light Source Estimation for Augmented Reality Using Deep Learning,” *The Visual Computer,* vol. 35, pp. 873-883.

[4] J. Caarls, P. Jonker, and S. Persa, “Sensor Fusion for Augmented Reality,” *Lecture Notes in Computer Science,* Springer.

[5] W. Fang, L. Zheng, H. Deng, and H. Zhang, “Real-Time Motion Tracking for Mobile Augmented Reality Using Visual-Inertial Fusion,” *Sensors,* vol. 17, no. 5.

[6] H. Chen et al., “High Energy Efficiency Mobile Augmented Reality Applications under Adaptive Object Detection Engine and Self-Learning Governor,” in *Proc. ACM SAC,* 2024.

[7] M. Alsweis and O. Deussen, “Procedural Techniques for Simulating the Growth of Plant Leaves and Adapting Venation Patterns,” in *Proc. ACM VRST,* 2015.

[8] J. Li et al., “Real-Time Physics-Based Interaction in Augmented Reality,” in *Proc. IEEE VR Conference,* 2022.

[9] H. Sang et al., “Scene Augmentation Methods for Interactive Embodied AI Tasks,” *IEEE Transactions on Instrumentation and Measurement,* 2023.

[10] E. Ragan, C. Wilkes, and D. A. Bowman, “Simulation of Augmented Reality Systems in Purely Virtual Environments,” in *Proc. IEEE Virtual Reality Conference*.

[11] H. Liu et al., “Real-Time Lighting Estimation for Augmented Reality via Differentiable Screen-Space Rendering,” *IEEE Transactions on Visualization and Computer Graphics,* 2022.

[12] M. Pluisch, J. Gugenheimer, and E. Kruijff, “Context-Aware Application Recommendations in Augmented Reality,” in *Proc. ACM Symposium on Spatial User Interaction,* 2025.

[13] W. Li, A. Y. C. Nee, and S. K. Ong, “A State-of-the-Art Review of Augmented Reality in Engineering Analysis and Simulation,” *Multimodal Technologies and Interaction,* vol. 1.

[14] M. Na and J. Lee, “Generative AI-Enabled Energy-Efficient Mobile Augmented Reality in Multi-Access Edge Computing,” *Applied Sciences*, 2024.

[15] M. Billinghurst, A. Clark, and G. Lee, “A Survey of Augmented Reality,” *Foundations and Trends in Human-Computer Interaction*, vol. 8.

11. APPENDICES

11.1 Appendix A – Survey Instruments

This appendix presents the instruments used to collect data during the evaluation of the proposed context-aware AR plant simulation system. Since the study involves **Grade 1 learners**, the data collection methods are kept simple and supported by teacher or facilitator observation**.**

The instruments include a structured observation sheet**, a** teacher feedback form**,** and a simple post-activity evaluation format. These tools will help record how students interact with the AR plant simulation, how they respond to environmental changes such as light and movement, and how easily they understand the virtual plant growth experience.

The collected information will be used to evaluate the usability, engagement, and effectiveness of the proposed AR learning system.

A.1 Observation Sheet

The observation sheet will be used during student interaction sessions to record behavior related to usability, engagement, environmental interaction, and understanding of plant growth concepts while using the AR plant simulation system.

Table 11.1: Observation Sheet

Observation Area	Description
System Interaction	Whether the student can successfully view and interact with the AR plant simulation using the mobile device.
Engagement Level	Whether the student remains interested and attentive while observing the virtual plant growth.
Environmental Interaction	Whether the student understands that environmental factors such as light or device movement affect the plant growth in the simulation.
Ease of Use	Whether the student can interact with the AR system without confusion or difficulty.
Learning Understanding	Whether the student shows basic understanding of how environmental conditions influence plant growth.
Task Completion	Whether the student successfully completes the activity using the AR plant simulation system.

A.2 Teacher / Facilitator Feedback Form

Sample statements:

• The AR storybook was easy for the child to use.

• The interaction design was appropriate for the age group.

• The AR content remained stable and understandable during use.

• The child appeared comfortable during the activity.

• The system supported learning and engagement effectively.

A.3 Post-Activity Evaluation

This will include a short teacher-assisted evaluation of learner response after the session, focusing on:

• level of participation

• understanding of the science concept

• ability to follow the interaction flow

• interest in repeating the activity

11.2 Appendix B – Detailed Diagrams

This appendix contains the detailed technical diagrams used to explain the structure and operation of the proposed system.

The appendix may include:

• Overall System Architecture Diagram

• Hybrid Interaction Framework Architecture Diagram

• Gantt Chart / Work Plan Diagram

B.1 Overall System Architecture

This diagram presents the complete AR storybook system with the four research components:

1. Hybrid Interaction Framework

2. Affective Visualization Model

3. Gamified Narrative Engine

4. Context-Aware Simulation Engine

B.2 Individual Component Diagram

This diagram shows the internal structure of the proposed **Context-Aware AR Plant Simulation System**, including the main modules that enable environmental sensing, plant growth simulation, and interactive feedback within the augmented reality environment.

The diagram includes the following components:

- Environmental Sensing Module
- Sensor Processing and Fusion Module
- Context-Aware Plant Growth Simulation Module
- Interactive Environmental Feedback Module
- AR Visualization and Interaction Module
- System Evaluation Module

11.3 Appendix C – Simulation Model and Parameter Design

This appendix presents the preliminary design of the plant growth simulation model and environmental parameter mapping used in the proposed context-aware AR simulation system**.** The diagrams illustrate how environmental inputs such as light intensity, device movement, and user interaction are translated into simulation parameters that influence the behavior of the virtual plant.

C.1 Planned Simulation Features

The proposed context-aware AR simulation system is expected to include the following core simulation features:

- **Environmental Light Detection** – The system detects real-world light intensity using mobile device sensors to determine environmental brightness conditions.

- **Context-Aware Plant Growth Simulation** – The virtual plant growth rate and development will change dynamically based on environmental parameters such as light exposure.

• **Motion-Based Environmental Effects** – Device motion detected through gyroscope sensors will simulate environmental effects such as wind interaction with the plant.

• **Flashlight-Based Virtual Sunlight** – The mobile device flashlight will act as an artificial sunlight source to support plant growth when the surrounding environment is too dark.

• **Real-Time Simulation Updates** – The plant simulation will continuously update in response to environmental changes and user interactions.

C.2 Purpose of Simulation Design

These preliminary simulation designs are included to demonstrate how the proposed system supports:

• Dynamic environmental interaction

• Real-time plant growth behavior
• Efficient mapping of sensor inputs into simulation parameters
• Improved realism of the virtual ecosystem within the AR environment

These simulation design elements illustrate how the proposed context-aware AR system integrates environmental sensing, procedural plant growth modelling, and interactive feedback to create a responsive virtual plant simulation.

11.4 Appendix D – Risk Analysis

This appendix presents the main risks associated with the development and evaluation of the proposed **context-aware AR plant simulation system**, along with the strategies planned to reduce or manage those risks.

Table 11.2: Risk Analysis

Risk	Description	Mitigation Strategy
Environmental Light Detection Errors	The system may incorrectly detect environmental lighting conditions, affecting plant growth simulation accuracy.	Use multiple sensor readings and calibration to improve light detection accuracy.
Sensor Data Instability	Inconsistent readings from sensors such as gyroscope or accelerometer may affect environmental interaction behavior.	Apply sensor fusion and smoothing techniques to stabilize sensor data.
AR Tracking Instability	The virtual plant may lose alignment with the real environment due to unstable AR tracking.	Use stable surface detection and continuous AR tracking updates during simulation.
Simulation Response Delay	The plant simulation may not respond quickly to environmental changes.	Optimize simulation algorithms and ensure efficient sensor data processing.
Device Performance Limitations	Some mobile devices may experience reduced performance when running the AR simulation.	Optimize rendering and simulation processes to reduce computational load.
Limited Testing Opportunities	Difficulty in conducting sufficient testing sessions with students or controlled environments.	Plan testing sessions in advance and prepare alternative testing environments.
Data Recording Issues	Problems in recording sensor logs or simulation results during testing.	Implement automatic logging and backup mechanisms for collected data.

11.5 Appendix E – Ethical Documents

This appendix includes the ethical and participant-related documents required for the research study, especially because the target participants are young children.

The appendix may include:

• Parent / Guardian Consent Form

• Child Assent Form

• Participant Information Sheet

• Privacy and Confidentiality Statement

• Ethical Clearance Submission Documents

E.1 Parent / Guardian Consent Form

This document will explain:

• the purpose of the study

• the child's involvement

• the type of activities to be carried out

• the voluntary nature of participation

• privacy protection measures

E.2 Child Assent Form

This will provide a simple explanation, in age-appropriate language, to confirm that the child is willing to participate.

E.3 Privacy and Confidentiality

All collected data will be anonymized. No unnecessary personal information will be stored, and all research data will be used only for academic purposes.